Progressive exercises

E1 · Direct application. In a basis of space, the vectors $\vec{u}$ and $\vec{v}$ have respective coordinates $(1;2;-1)$ and $(3;6;-3)$. Are they collinear? Justify.

E2 · Application and justification. In an orthonormal coordinate system of space, consider $A(0;1;2)$ and $B(4;-1;0)$. Determine the coordinates of the midpoint of $[AB]$, then calculate the distance AB.

E3 · Method to organise. In a basis of space, consider $\vec{u} = (1;0;1)$, $\vec{v} = (0;1;1)$ and $\vec{w} = (2;3;5)$. Express $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$, then deduce that these three vectors are coplanar.

E4 · Choice of strategy. In a basis of space, the vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ have coordinates $(1;0;1)$, $(0;1;1)$ and $(1;1;0)$. Is the triple $(\vec{u},\vec{v},\vec{w})$ a basis of space? Justify.

E5 · Synthesis or proof. In a coordinate system of space, consider $A(1;0;2)$, $B(3;-1;4)$ and $C(5;-2;6)$. Determine whether C lies on the line (AB). Justify by vector calculation.